

Charts: Predicting The Success Of A Song Using Metrics From Spotify

Manan Gupta, Priyanshi Jain, Aarush Talwar
Indraprastha Institute of Information Technology - Delhi
{manan17372, priyanshi17358, aarush17213}@iiitd.ac.in

Abstract

In this project, we aim to predict which songs will become Billboard Hot 100 hits. We created a dataset of around 8000+ songs from Spotify, comprising of hits and non-hits, using Spotify Web API and extracted song audio features for the same.

1. INTRODUCTION

The Billboard Hot 100 Chart is highly acclaimed and is one of the most popular ways to measure the success of a song. Using Machine Learning techniques on song audio features extracted from the Spotify Web API, we predict whether or not a song will be a Billboard Hot 100 hit. Our ML algorithms output a binary prediction of whether or not the song will feature on the Billboard Hot 100 list. Our work will be relevant to artists and producers. This study will help artists better understand what features of their music could help maximize their chance of success, and could also help music labels to predict which songs have the best returns on investment on advertising.

2. RELATED WORK

Music technology companies like The Echo Nest and Next Big Sound have been using data analytics to help artists and labels predict and track a song's success for almost a decade. This problem is referred to as Hit Song Science (HSS) in the Music Information Retrieval(MIR) field.

The HSS problem is often researched with the help of machine learning tools. Researchers have used Convolutional Neural Networks [1] and K-Means Clustering [2] to predict pop hits. These studies focused more heavily on the signal-processing involved in audio analysis and were reasonably successful.

Another research group used SVM to predict the top 10 dance hits [3]. Another study tried to classify songs based on lyric content [4]. Though the review was reasonably successful at ranking many singles, it had a lot of false negatives. It thus concluded that the analysis of lyrics was not an effective method for this problem.

3. METHODOLOGY

3.1 Data and Evaluation

The data was extracted using the Spotify Web API. We used the search function to obtain a list of songs from each year between 1990-2018, let's call this set s1. After this, we found the Billboard Hot 100 playlist for each of these years using the search function of the API again and then created another set with all of the songs from these playlists, let's call this set s2. Having done this, we removed all duplicate elements of s2 in s1. Now, we queried the Spotify API for the song attributes of each song, and after adding labels for the hit and non-hit songs, a CSV file can be generated. After experimenting with the number of elements in s1 and s2, we generated a balanced data set with around 2200 hits and 2200 non-hits. Another unbalanced dataset was also obtained in which there were 2200 ~ hits and 6000 ~ non-hits. We experimented with oversampling the unbalanced dataset to obtain a new pseudo-balanced dataset with about 12,000 samples.

3.1.1 Extracted Features

The CSV file contains the following attributes for each song: 'danceability', 'energy', 'loudness', 'speechiness', 'acousticness', 'instrumentalness', 'liveness', 'valence', 'tempo', 'duration_ms', 'popularity', 'billboard', 'year'. 'Billboard' is the binary classifier that separates the hits and non-hits. We added two more features: 'Artist popularity' and 'Artist score'. 'Artist popularity' was obtained using Spotify Web API. 'Artist score' is a binary metric with a value of 1 if the artist had a previous hit and 0 if not. For this feature, we checked the hits till 1976. Considering the average career spans of artists, we feel that we have covered all possibilities, i.e. it is highly unlikely that an artist had a hit song before 1976 and after 1990.

3.1.2 Evaluation Metrics

Accuracy, precision, recall are being used for evaluation of our model. Accuracy is not a good measure for the

evaluation of an unbalanced dataset, so we are using precision and recall.

3.2 Algorithms

To predict if a song will make it to Billboard or not we employed four algorithms: Logistic Regression, SVM, Random Forests and Neural Networks, since all of these algorithms are commonly used for binary classification tasks. We used 3-folds cross-validation procedure due to the limited size of our dataset.

4. RESULT AND ANALYSIS

4.1 Dataset

We generated a balanced dataset with ~2000 hits and ~2000 non-hit songs. Suspecting that the dataset was too sparse, we generated another dataset with ~2000 hits and ~6000 non-hit songs. We also over sampled the positive samples of the balanced dataset to form another dataset of ~12000 samples. The oversampled dataset showed a tendency to overfit. Meanwhile the unbalanced dataset showed good accuracy but low precision for the positive class. As a result, we decided to use the balanced dataset with ~2000 hits and ~2000 non-hit songs for further computations.

To gain an initial insight into the dataset, we constructed scatter plots of the two classes, considering two features at a time. Scatter plots for some of the ‘strong’ features (explained in section 4.2) are given below.

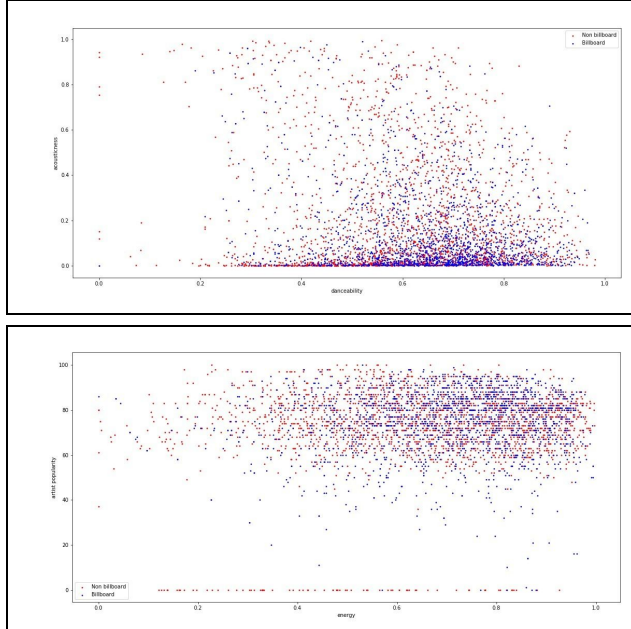


Figure 1: Scatter plots for classes with ‘strong’ features

4.2 Features

We used the following features: ‘danceability’, ‘energy’, ‘loudness’, ‘speechiness’, ‘acousticness’, ‘liveness’, ‘valence’, ‘tempo’, ‘billboard’, ‘year’ and ‘Artist popularity’ and ‘Artist score’. ‘Billboard’ is the binary classifier that separates the hits and non-hits. We added two more features:

‘Year’ was one-hot encoded to avoid giving more unfair weightage to later years. The feature ‘instrumentalness’ wasn’t used due to redundancy and missing points in data. Accuracy estimation according to individual features using Logistic Regression yielded that the ‘artist score’ gives the highest accuracy. Other features which performed well were: ‘danceability’, ‘energy’, ‘loudness’, ‘acousticness’. When the same analysis is done on batches of 7 years, the accuracy almost all features increases while the accuracy for some of the features are almost the same.

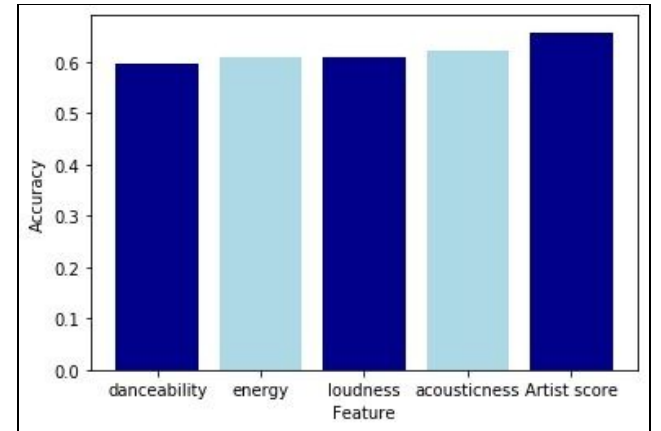


Figure 2: Accuracy estimation according to individual features using Logistic Regression. (Only the top features are plotted)

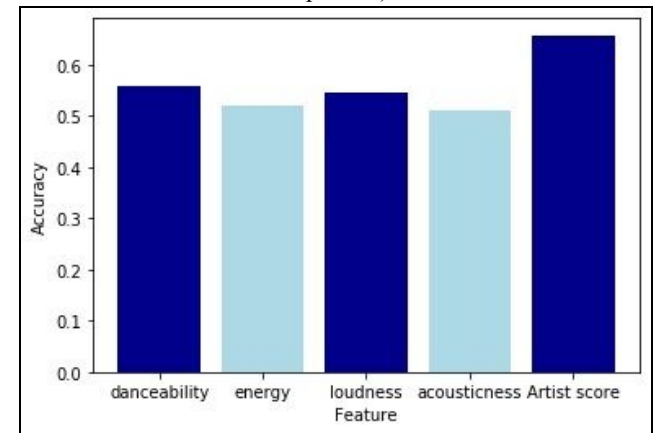


Figure 3: Accuracy Estimation according to individual features using Logistic Regression in batches of 7 years. (Only the top features are plotted)

4.3 Algorithms

Due to sparsity of the dataset, 3-fold cross validation is used and the average accuracy, precision and recall values are reported. The performances of all models are summarized in table 1

4.3.1 Logistic Regression:

Very minimal variance between training and validation sets was observed using Logistic Regression. An accuracy of about 0.66, precision of about 0.65 for both classes, and recall close to 0.50 was observed. LR with L2 regularization saw a slight gain in recall, the other metrics were almost the same. Logistic regression performs considerably well for a baseline model, compared to other baseline models such as random predictors.

4.3.2 Support Vector Machine:

Considering that the data is not linearly separable (Figure 1 verifies the claim), we explore various kernels to classify the data. Thus, SVM with a linear kernel shows the least impressive performance. Although polynomial kernels provide a gain in accuracy, results are unreliable as they are usually biased towards one or the other class. This can be confirmed from the precision values for the two classes. The RBF kernel provides one of the best performances among all models. The model shows good consistency across training and validation sets with an accuracy of about 0.74. We note precision of about 0.7 for both classes and one of the highest recalls among all models.

4.3.3 Random Forests:

Random Forests comes out to be one of the most successful algorithms. It yields an accuracy of around 70% for the validation set and a high precision value of around 0.8 for class 1.

For Random Forests, the maximum depth of the forest is selected to be 6 using the below graph. After 6, we can see that there is considerable overfitting.

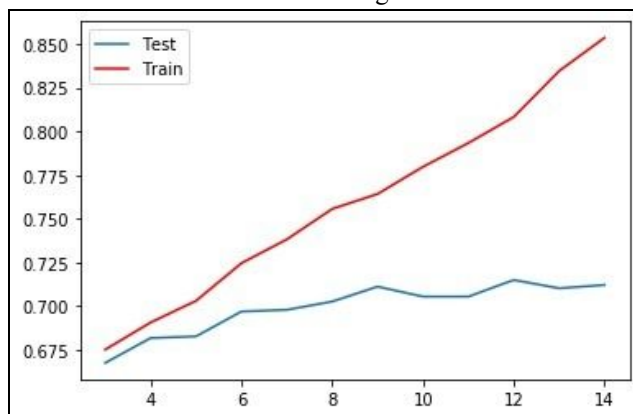


Figure 4: Test and train Accuracies vs maximum depth of the forest for Random Forest

4.3.4 Neural Networks:

Neural network doesn't prove to be very fruitful for this problem. As we increase the hidden layers, the accuracy was decreasing implying that the data isn't set for neural networks, probably because of the small size of the dataset. The best accuracy was achieved with one hidden layer with 6 nodes at 10,000 epochs. With more epochs, the data tends to overfit.

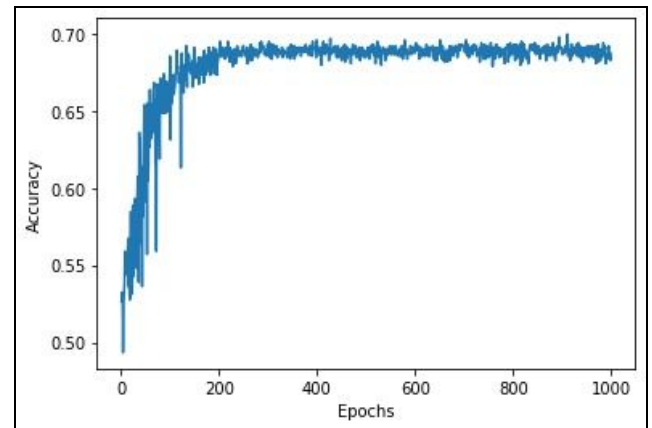


Figure 5: Accuracy vs Epochs for NN.

4.4 Batch Wise analysis

We explore the hypothesis that trends in popular music change with time. Thus, we carry out all analyses restricted to a span of 7 years. After this restriction, almost features become stronger indicators of popularity. Figure 3 verifies this claim. We note an increase in the performance of SVM for more recent years, indicating that trends in popular music have shown a tendency to solidify in recent times. Other models did not show such gains in performance, this might be attributed to the sparsity of datasets after the split. Some models showed a dip in performance in the 1991-1997 period.

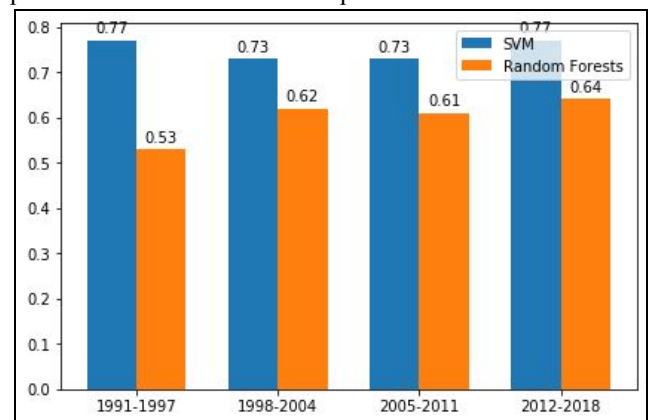


Figure 6: Batch wise accuracy for SVM and RF models.

	Accuracy		Precision (1)		Precision (0)		Recall	
	TRAIN	VAL	TRAIN	VAL	TRAIN	VAL	TRAIN	VAL
LR	0.6623	0.6545	0.6759	0.6491	0.6549	0.6669	0.5142	0.5066
LR with L2	0.6588	0.6586	0.669	0.6532	0.6534	0.6697	0.5147	0.5124
SVM RBF	0.7585	0.7389	0.7543	0.6901	0.7284	0.697	0.5578	0.5242
SVM Linear	0.6438	0.6438	0.6669	0.6334	0.6333	0.6669	0.4543	0.4543
SVM Polynomial 2	0.6789	0.6686	0.7093	0.6557	0.6637	0.6941	0.5119	0.4989
SVM Polynomial 3	0.7165	0.6769	0.7581	0.6643	0.4647	0.701	0.5633	0.5204
Random Forest	0.7319	0.6905	0.7811	0.7277	0.7063	0.6718	0.5813	0.5252
Neural Network	0.733	0.6874	0.7721	0.6739	0.712	0.7142	0.5965	0.5387

Table 1: Evaluation Matrix for Various Models

5. CONCLUSION AND FUTURE WORK

From the results encapsulated in Table 1, we note that random Forests and SVM with RBF kernel perform the best on the dataset with the highest accuracy, precision and recall values. These models show minimal variance and consistent results across training and validation sets. SVM shows no signs of overfitting, restricting the depth of trees in random forests solves the overfitting problem there. In the future, we hope to work with more data by including songs from more years. We feel that clustering data according to year may lead to better results. Including encoded information about song genre may also lead to better results. In the best case scenario, one might be able to predict the approximate rank of a song on the Billboard charts.

6. REFERENCES

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